

Операционные системы

Командная оболочка Midnight Commander

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Цели и задачи работы

Процесс выполнения лабораторной работы

Выводы по проделанной работе

1. Цели и задачи работы

Ознакомление с файловой системой Linux, её структурой, именами и содержанием каталогов. Приобретение практических навыков по применению команд для работы с файлами и каталогами, по управлению процессами, по проверке использования диска и обслуживанию файловой системы.

- 1 Изучить возможности Midnight Commander
- 2 Изучить редактор Midnight Commander

2. Процесс выполнения лабораторной работы

Работа с Midnight Commander

The screenshot shows the Midnight Commander (mc) interface. At the top, the title bar reads "mc [miradov@miradov]:~ — /usr/bin/mc -P /tmp/mc.pwd.AwTtY". Below it, the address bar shows "miradov@miradov:~/work/study/2025-2026/OneP" and "mc [miradov@miradov]:~ — /usr/bin/mc -P /tmp". The main area is divided into two panels, "Left" and "Right", each displaying a directory listing. The "Left" panel shows the current directory as "~" and the "Right" panel shows the current directory as "~". Both panels display a table of files and directories with columns for Name, Size, Modify, and time. The "Left" panel shows files like .cache, .config, .gnupg, .local, .mozilla, .ssh, .texlive2025, Desktop, Documents, Downloads, Music, Pictures, Public, Templates, Videos, .australia, .git-extended, .monthly, .play, .reports, .ski.plases, .work, .bash_history, .bash_logout, .bash_profile, .bashrc, .emacs, and .gitconfig. The "Right" panel shows the same files and directories. At the bottom, a status bar shows "UP--DIR" and "866 / 1486 (58%)". Below the status bar, a hint says "Hint: Tab changes your current panel." and a prompt "miradov@miradov:~\$". At the very bottom, a menu bar shows "1Help 2Menu 3View 4Edit 5Copy 6RenMov 7Mkdir 8Delete 9PullDn10Quit".

Left	File	Command	Options	Right			
~	Name	Size	Modify time	~	Name	Size	Modify time
UP--DIR	Aug 31 09:52			UP--DIR	Aug 31 09:52		
.cache	488	Aug 31 11:23		.cache	488	Aug 31 11:23	
.config	364	Aug 31 11:23		.config	364	Aug 31 11:23	
.gnupg	136	Aug 31 10:02		.gnupg	136	Aug 31 10:02	
.local	20	Aug 31 09:53		.local	20	Aug 31 09:53	
.mozilla	34	Oct 23 2025		.mozilla	34	Oct 23 2025	
.ssh	132	Aug 31 10:04		.ssh	132	Aug 31 10:04	
.texlive2025	18	Aug 31 10:07		.texlive2025	18	Aug 31 10:07	
Desktop	0	Aug 31 09:53		Desktop	0	Aug 31 09:53	
Documents	0	Aug 31 09:53		Documents	0	Aug 31 09:53	
Downloads	0	Aug 31 09:53		Downloads	0	Aug 31 09:53	
Music	0	Aug 31 09:53		Music	0	Aug 31 09:53	
Pictures	0	Aug 31 09:53		Pictures	0	Aug 31 09:53	
Public	0	Aug 31 09:53		Public	0	Aug 31 09:53	
Templates	0	Aug 31 09:53		Templates	0	Aug 31 09:53	
Videos	0	Aug 31 09:53		Videos	0	Aug 31 09:53	
.australia	0	Aug 31 11:07		.australia	0	Aug 31 11:07	
.git-extended	74	Aug 31 10:31		.git-extended	74	Aug 31 10:31	
.monthly	24	Aug 31 11:03		.monthly	24	Aug 31 11:03	
.play	26	Aug 31 11:08		.play	26	Aug 31 11:08	
.reports	14	Aug 31 11:03		.reports	14	Aug 31 11:03	
.ski.plases	28	Aug 31 11:06		.ski.plases	28	Aug 31 11:06	
.work	10	Aug 31 10:03		.work	10	Aug 31 10:03	
.bash_history	2116	Aug 31 11:23		.bash_history	2116	Aug 31 11:23	
.bash_logout	18	Jul 23 2025		.bash_logout	18	Jul 23 2025	
.bash_profile	144	Jul 23 2025		.bash_profile	144	Jul 23 2025	
.bashrc	677	Aug 31 10:25		.bashrc	677	Aug 31 10:25	
.emacs	334	Nov 14 2025		.emacs	334	Nov 14 2025	
.gitconfig	170	Aug 31 10:34		.gitconfig	170	Aug 31 10:34	

UP--DIR 866 / 1486 (58%)

Hint: Tab changes your current panel.

miradov@miradov:~\$

1Help 2Menu 3View 4Edit 5Copy 6RenMov 7Mkdir 8Delete 9PullDn10Quit

Рисунок 1: Запуск mc

Работа с Midnight Commander

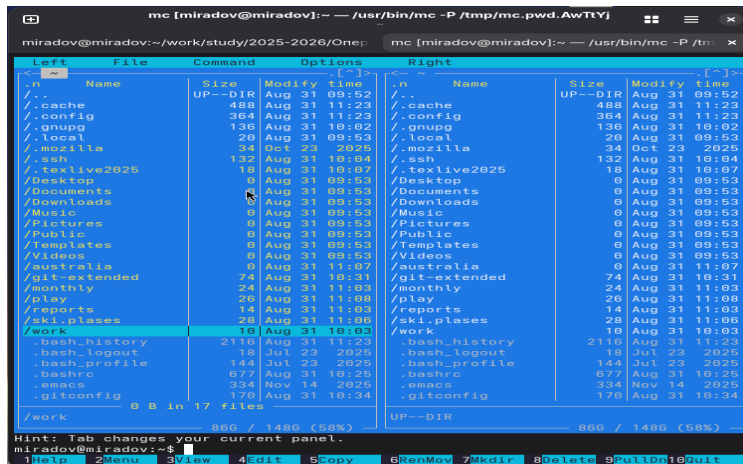


Рисунок 2: Выделение

Работа с Midnight Commander

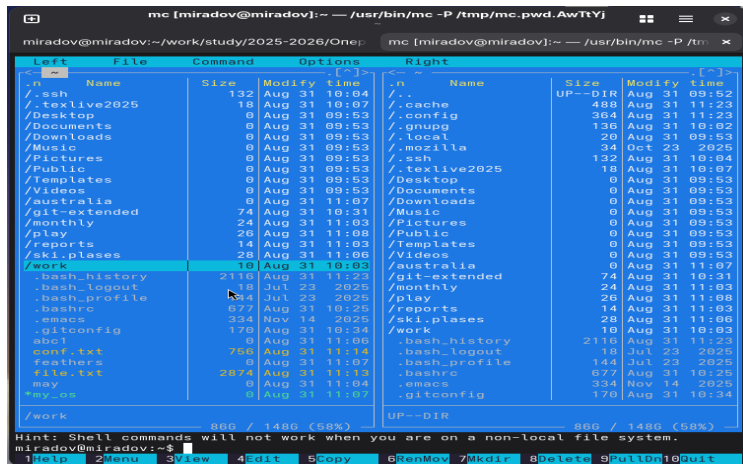


Рисунок 3: Отмена

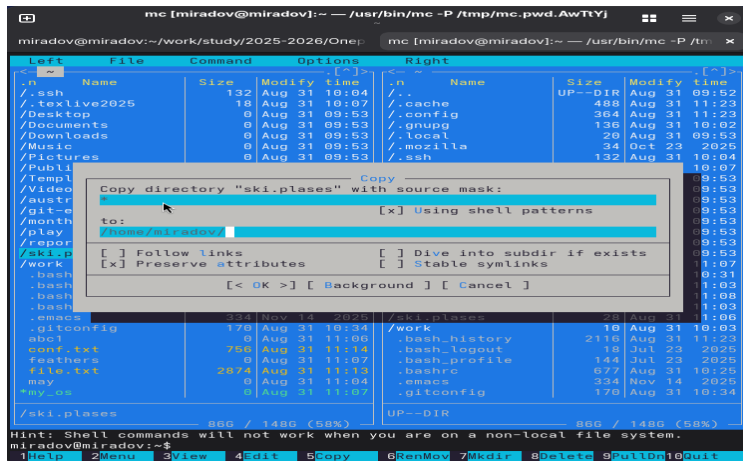


Рисунок 4: Копирование

Работа с Midnight Commander

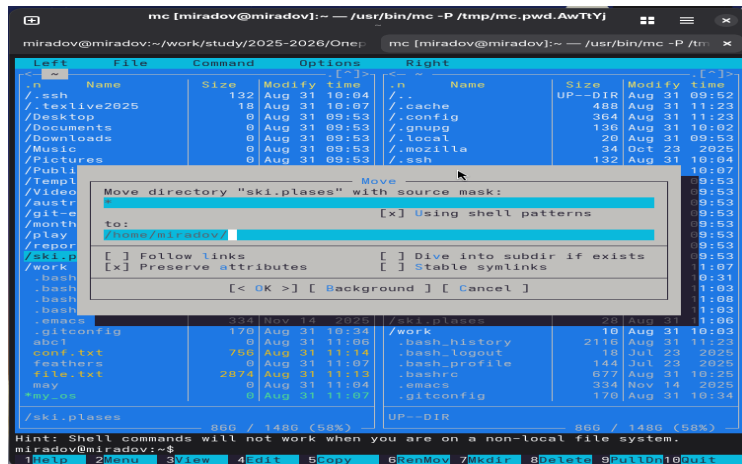


Рисунок 5: Перемещение

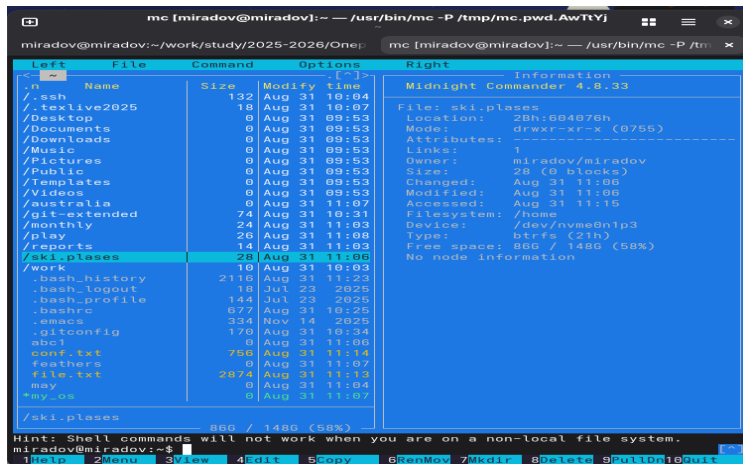


Рисунок 6: Информация

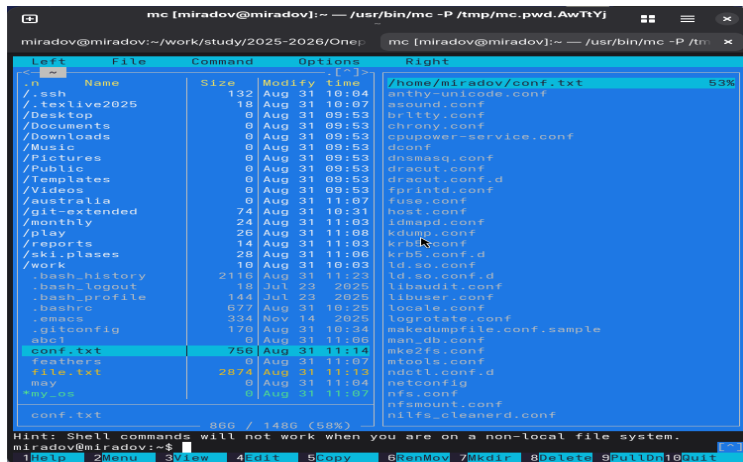


Рисунок 7: Быстрый просмотр

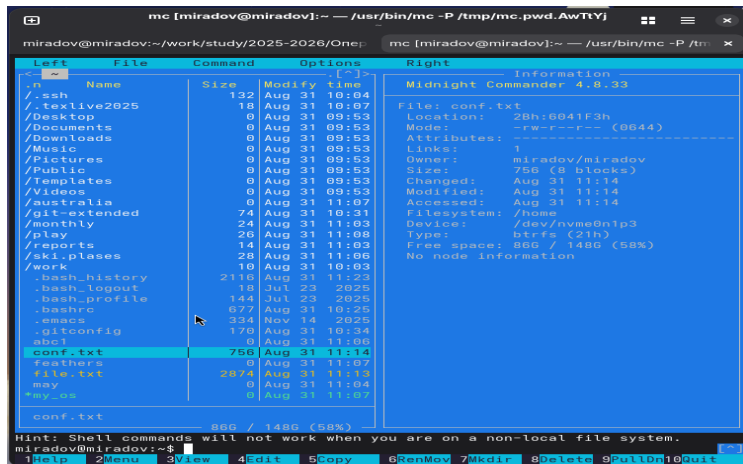


Рисунок 8: Информация

Работа с Midnight Commander

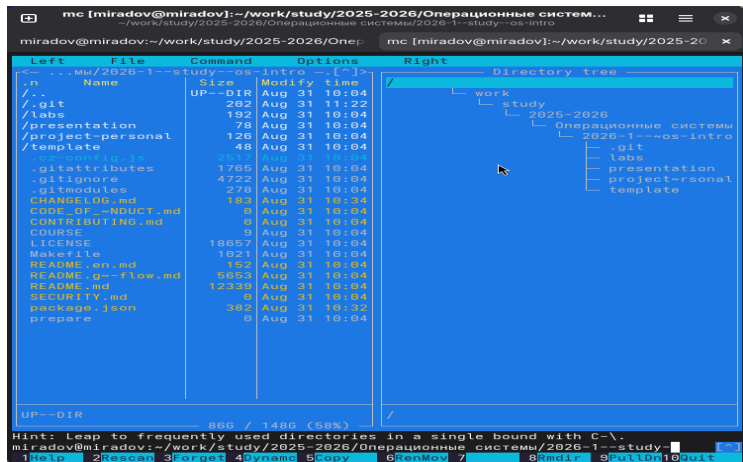
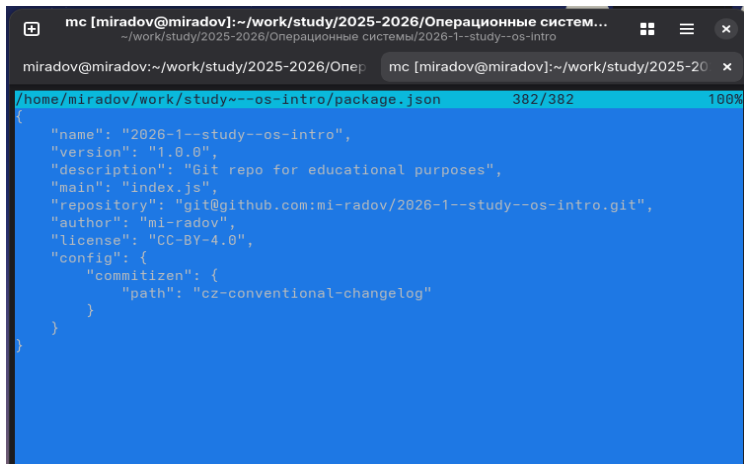


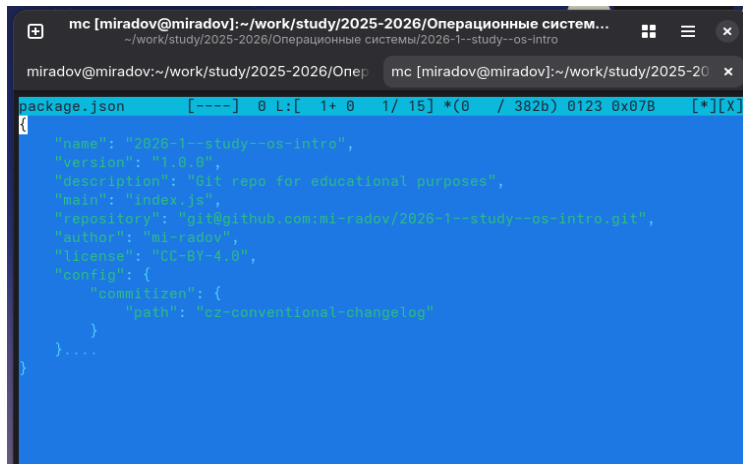
Рисунок 9: Дерево каталогов



The screenshot shows the Midnight Commander (mc) interface. The top bar displays the current directory: `~/work/study/2025-2026/Операционные системы/2026-1--study--os-Intro`. The left pane shows the file `package.json` selected, with its size `382/382` and a zoom level of `100%`. The right pane displays the JSON content of the file:

```
{
  "name": "2026-1--study--os-intro",
  "version": "1.0.0",
  "description": "Git repo for educational purposes",
  "main": "index.js",
  "repository": "git@github.com:mi-radov/2026-1--study--os-intro.git",
  "author": "mi-radov",
  "license": "CC-BY-4.0",
  "config": {
    "commitizen": {
      "path": "cz-conventional-changelog"
    }
  }
}
```

Рисунок 10: Просмотр содержимого текстового файла



The image shows a screenshot of the Midnight Commander (MC) file manager and editor. The title bar at the top reads "mc [miradov@miradov]:~/work/study/2025-2026/Операционные систем...". Below the title bar, the current directory is shown as "~/work/study/2025-2026/Операционные системы/2026-1--study--os-intro". The main window displays the contents of a file named "package.json". The file's content is as follows:

```
{
  "name": "2026-1--study--os-intro",
  "version": "1.0.0",
  "description": "Git repo for educational purposes",
  "main": "index.js",
  "repository": "git@github.com:mi-radov/2026-1--study--os-intro.git",
  "author": "mi-radov",
  "license": "CC-BY-4.0",
  "config": {
    "commitizen": {
      "path": "cz-conventional-changelog"
    }
  }
}
```

Рисунок 11: Отредактируем содержимое текстового файла без сохранения результатов

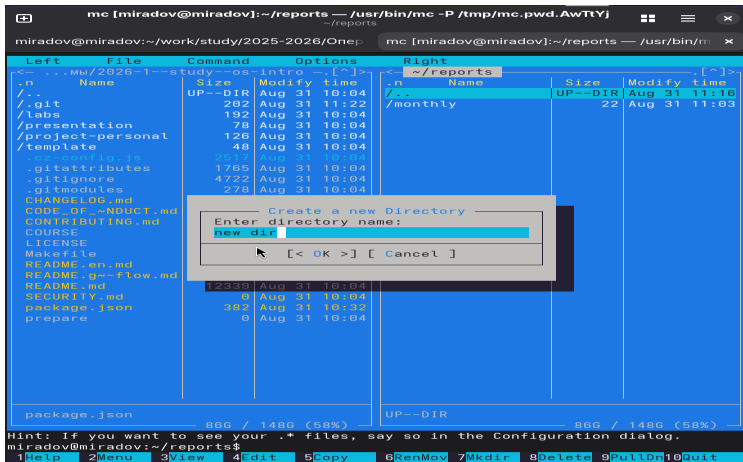


Рисунок 12: Создание каталога

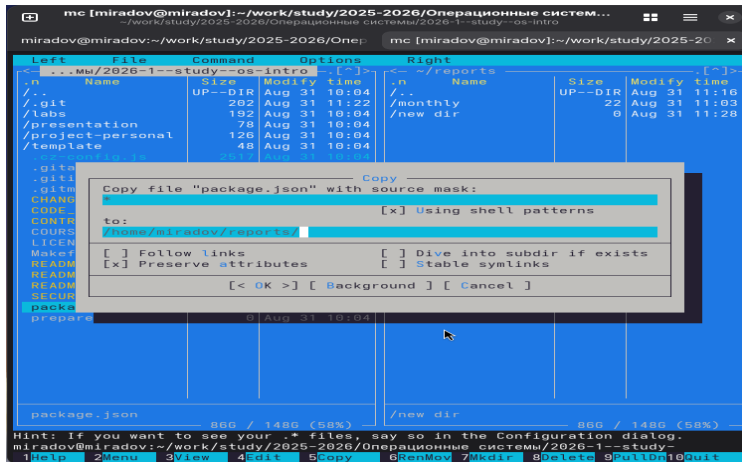


Рисунок 13: Копирование в файлов в созданный каталог

Работа с Midnight Commander

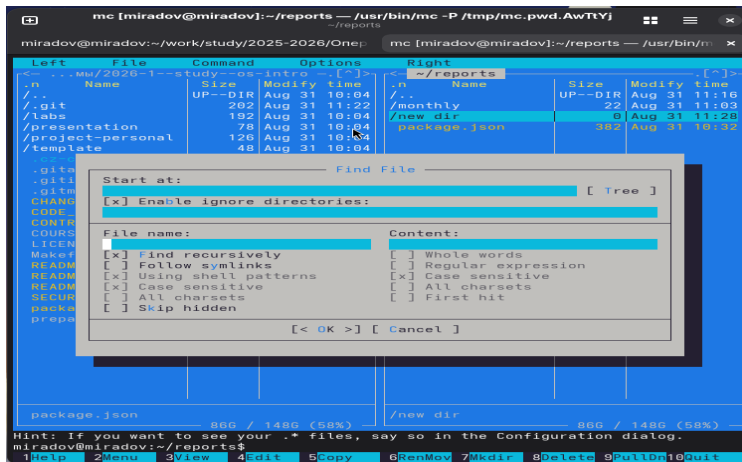


Рисунок 14: Поиск файлов

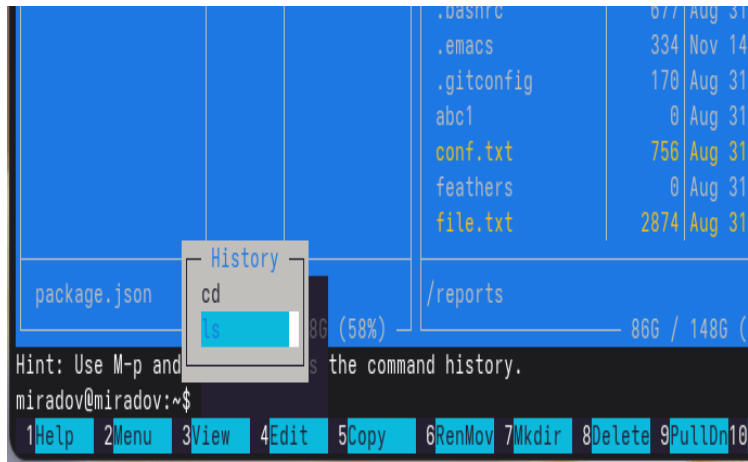


Рисунок 15: История команд

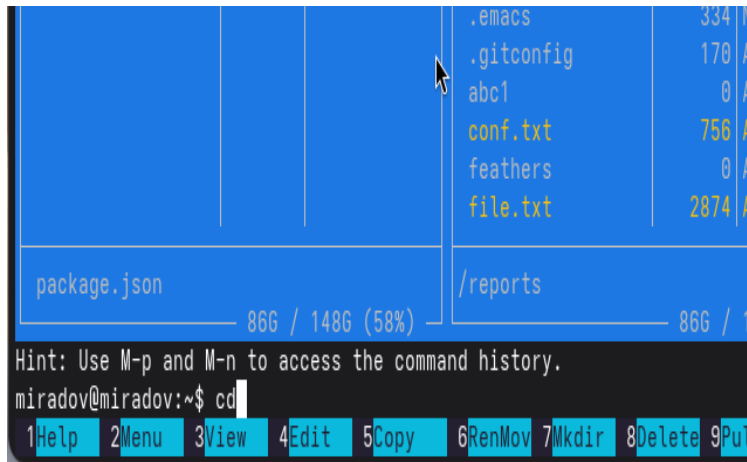
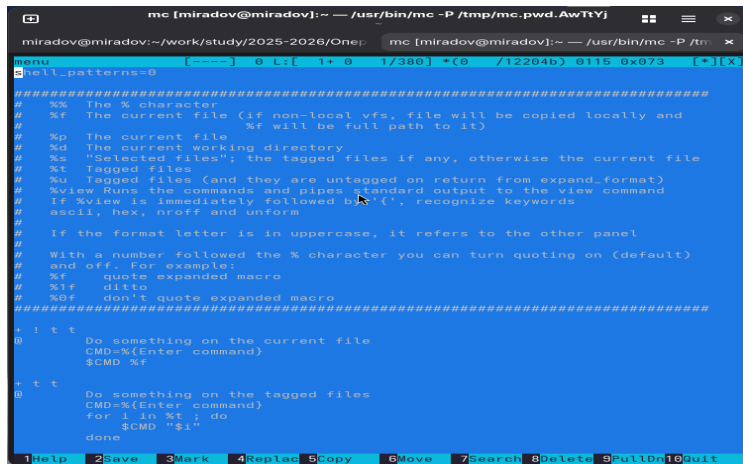


Рисунок 16: Переход в домашний каталог



```
mc [miradov@miradov]:~ — /usr/bin/mc -P /tmp/mc.pwd.AwTTY]
miradov@miradov:~/work/study/2025-2026/OneP: mc [miradov@miradov]:~ — /usr/bin/mc -P /tmp/mc.pwd.AwTTY]
mc.ext.ini [-----] 0 L: [ 1+ 0 1/1183] +(0 /28366b) 0035 8x023 [+] [X]
# Midnight Commander 4.0 extension file
#
# Warning: The structure of this file has been completely changed with the version
# 4.0. All lines starting with # or empty lines are ignored.
#
# IMPORTANT: mc scans this file only upon first use or after editing it using the
# mc "Edit extension file" command (F9-c-e). If you edit this file in any other way
# while mc is running, you will need to press F9-c-e and exit the editor for your
# changes to take effect, or exit mc and start it again.
#
# Section name can be anything with following exceptions:
#   there are two reserved section names:
#     mc.ext.ini
#     Default
#   special name pattern:
#     include/xxxxx
# See below for more details.
#
# Section [mc.ext.ini] is mandatory. It contains file metadata.
# "Version" parameter is mandatory. It contains the file format version.
#
# Section [Default] is optional. It is applied only if no other match was found.
#
# Sections like [include/xxxx] can be referenced as "include=xxxx" from other sect
# Section [include/xxxx] can be located before or after sections that point to it.
#
# Sections are processed from top to bottom, thus the order is important.
# Multiple sections with the same name are allowed, they are merged together.
# Sections may contain the same key multiple times; the last entry wins.
#
# [Default] should be a catch-all action and come last.
#
# A section describing a file can contain following keys:
#
#   File descriptions:
# 1 Help 2 Save 3 Mark 4 Replac 5 Copy 6 Move 7 Search 8 Delete 9 PullDn 10 Quit
```

Рисунок 17: Просмотр файла расширений

A screenshot of the Midnight Commander (mc) application's file menu. The window title is 'mc [miradov@miradov]:~ — /usr/bin/mc -P /tmp/mc.pwd.AwTtY'. The menu is open, showing various options and their descriptions. The background is blue, and the text is white. The menu options are listed at the bottom of the window.

```
mc [miradov@miradov]:~ — /usr/bin/mc -P /tmp/mc.pwd.AwTtY
miradov@miradov:~/work/study/2025-2026/OneP mc [miradov@miradov]:~ — /usr/bin/mc -P /tm
menu [----] 0 L: [ 1+ 0 1/380] *(0 /12204b) 0115 0x073 [+][X]
hell_patterns=0
#####
# %% The % character
# %f The current file (if non-local vfs, file will be copied locally and
#    %f will be full path to it)
# %p The current file
# %d The current working directory
# %s "Selected files"; the tagged files if any, otherwise the current file
# %t Tagged files
# %u Tagged files (and they are untagged on return from expand_format)
# %view Runs the commands and pipes standard output to the view command
# If %view is immediately followed by '(', recognize keywords
#    ascii, hex, nroff and unform
#
# If the format letter is in uppercase, it refers to the other panel
#
# With a number followed the % character you can turn quoting on (default)
# and off. For example:
# %f quote expanded macro
# %if ditto
# %of don't quote expanded macro
#####
+ ! t t
@ Do something on the current file
  CMD=%{Enter command}
  $CMD %f
+ t t
@ Do something on the tagged files
  CMD=%{Enter command}
  for i in %t ; do
    $CMD "$i"
  done
1Help 2Save 3Mark 4Replac 5Copy 6Move 7Search 8Delete 9PullDn10Quit
```

Рисунок 18: Просмотр файла меню

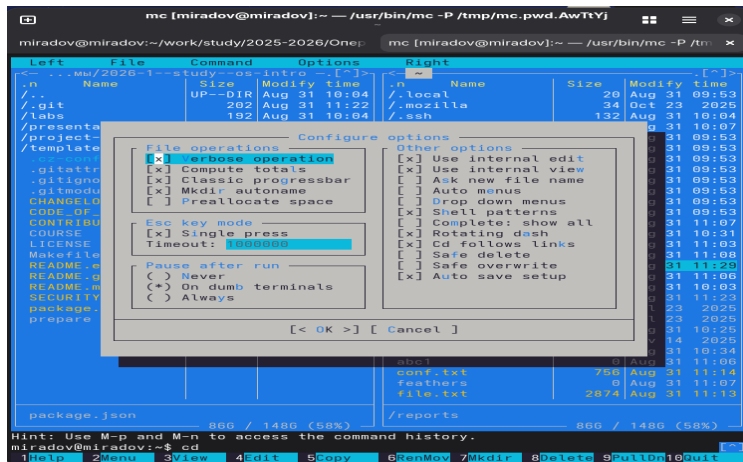


Рисунок 19: Конфигурация

Работа с Midnight Commander

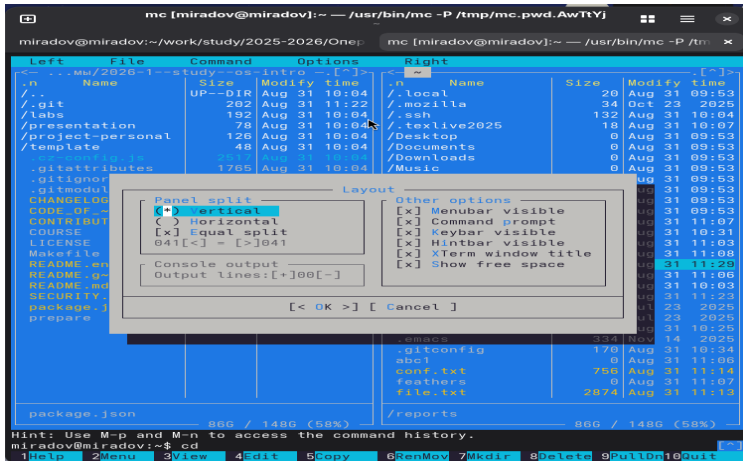


Рисунок 20: Внешний вид

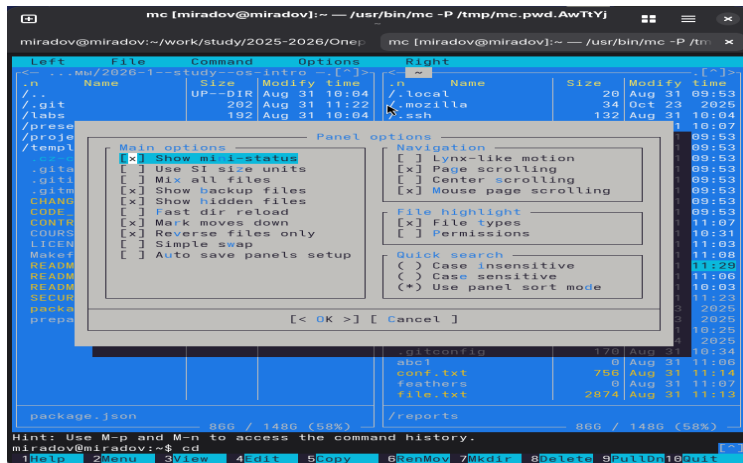


Рисунок 21: Настройки панелей

Работа с Midnight Commander

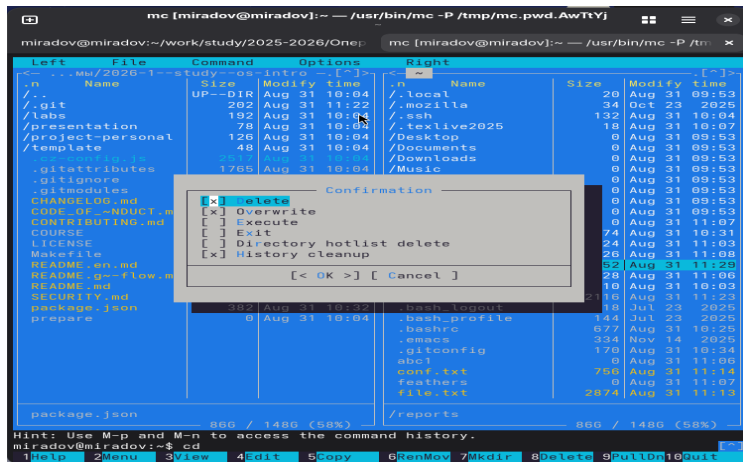


Рисунок 22: Подтверждение

Работа с Midnight Commander

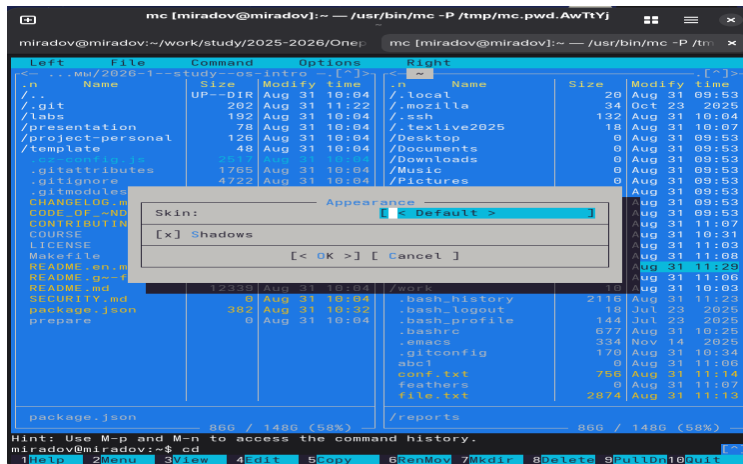


Рисунок 23: Оформление

Работа с Midnight Commander

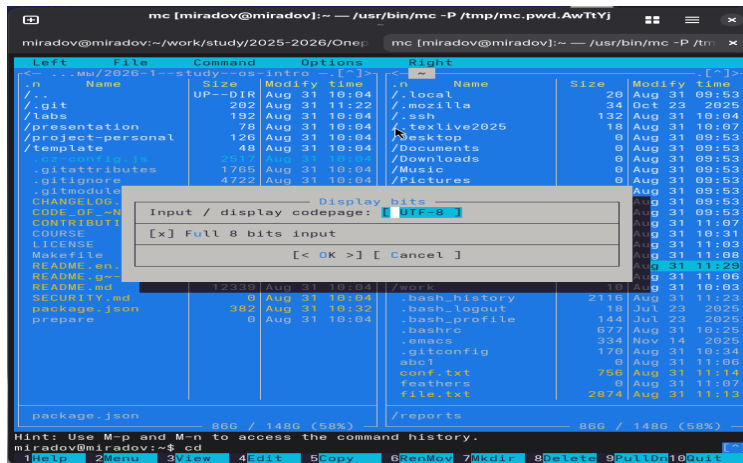


Рисунок 24: Кодировка символов

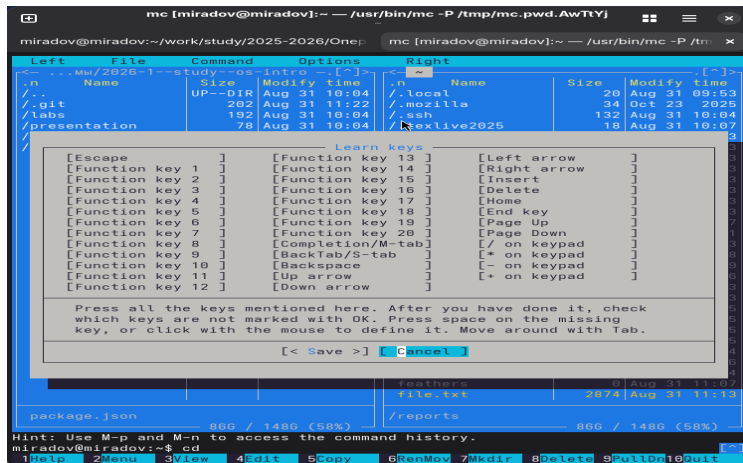
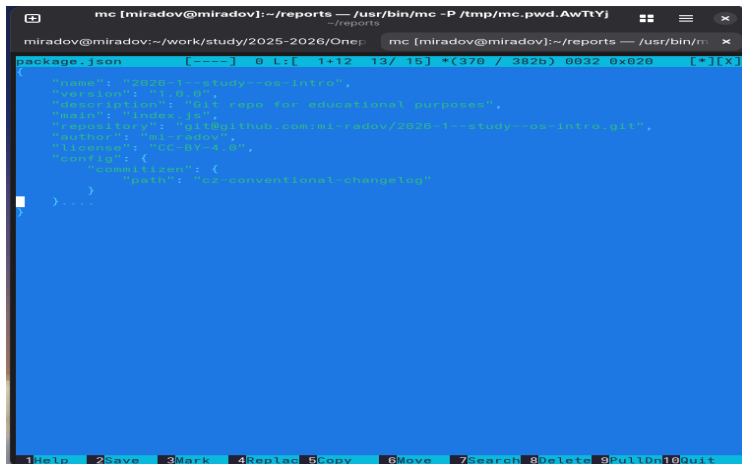


Рисунок 25: Распознавание клавиш

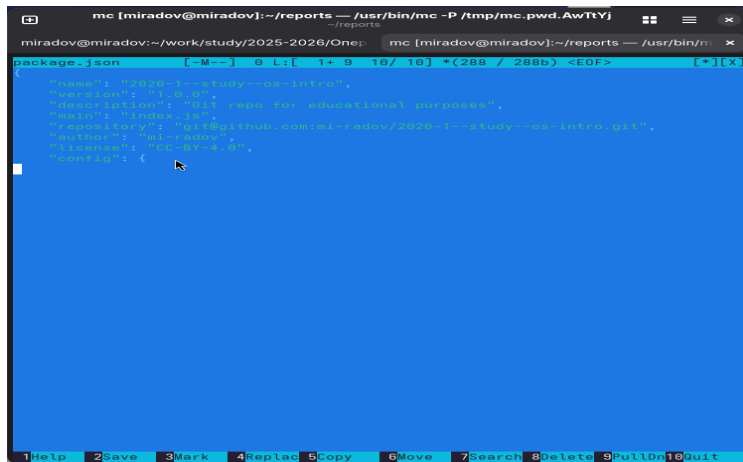


The screenshot shows the Midnight Commander (mc) file editor interface. The title bar indicates the current directory is `~/reports` and the file being edited is `package.json`. The editor displays the content of `package.json`, which is a JSON object for a project named `2026-1--study--ee-intro`. The JSON content is as follows:

```
{
  "name": "2026-1--study--ee-intro",
  "version": "1.0.0",
  "description": "Git repo for educational purposes",
  "main": "index.js",
  "repository": "git@github.com:mi-radov/2026-1--study--ee-intro.git",
  "author": "mi-radov",
  "license": "CC-BY-4.0",
  "config": {
    "conventional": {
      "path": "cz-conventional-changelog"
    }
  },
  ...
}
```

At the bottom of the window, a menu bar lists various actions: 1 Help, 2 Save, 3 Mark, 4 Replac, 5 Copy, 6 Move, 7 Search, 8 Delete, 9 PullDn, 10 Quit.

Рисунок 26: Файл с текстом



The screenshot shows the Midnight Commander (MC) interface. The title bar indicates the current directory is `~/reports` and the file being edited is `package.json`. The file content is displayed in a blue background with white text. The file is a JSON object with the following fields: `name`, `version`, `description`, `main`, `repository`, `author`, and `license`. The `config` field is an empty object. The bottom status bar shows various keyboard shortcuts for editing and file management.

```
mc [miradov@miradov]:~/reports — /usr/bin/mc -P /tmp/mc.pwd.AwTtYj
miradov@miradov:~/work/study/2025-2026/Onep: mc [miradov@miradov]:~/reports — /usr/bin/m
package.json [-M--] 0 L: [ 1+ 9 10/ 10] *(288 / 288b) <E0F> [*][X]
{
  "name": "2026-1--study--ee-intro",
  "version": "1.0.0",
  "description": "Git repo for educational purposes",
  "main": "index.js",
  "repository": "git@github.com:mi-radov/2026-1--study--ee-intro.git",
  "author": "mi-radov",
  "license": "CC-BY-4.0",
  "config": {
  }
}
```

1 Help 2 Save 3 Mark 4 Replac 5 Copy 6 Move 7 Search 8 Delete 9 PullDn 10 Quit

Рисунок 27: Файл с текстом

Работа с редактором Midnight Commander

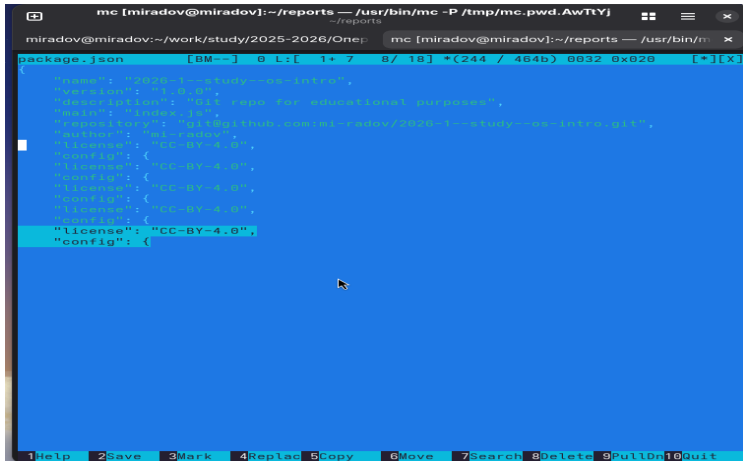


Рисунок 28: Копирование фрагмента

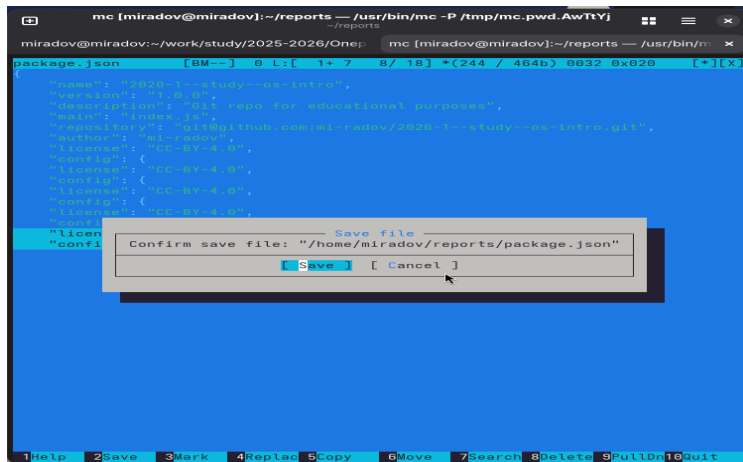


Рисунок 29: Сохранение

Работа с редактором Midnight Commander

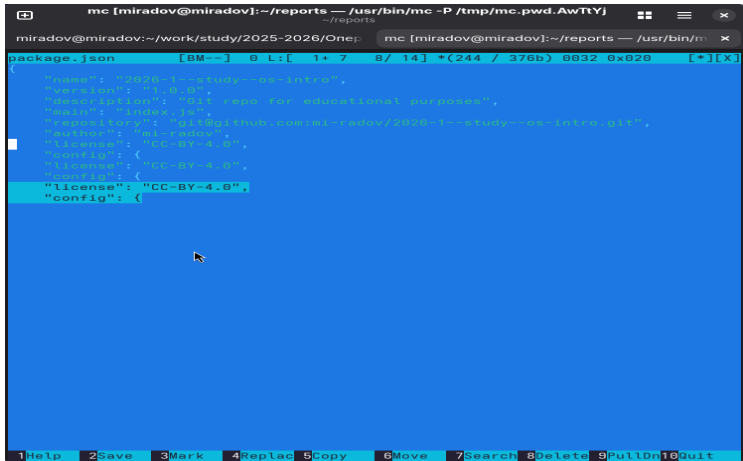


Рисунок 30: Отмена

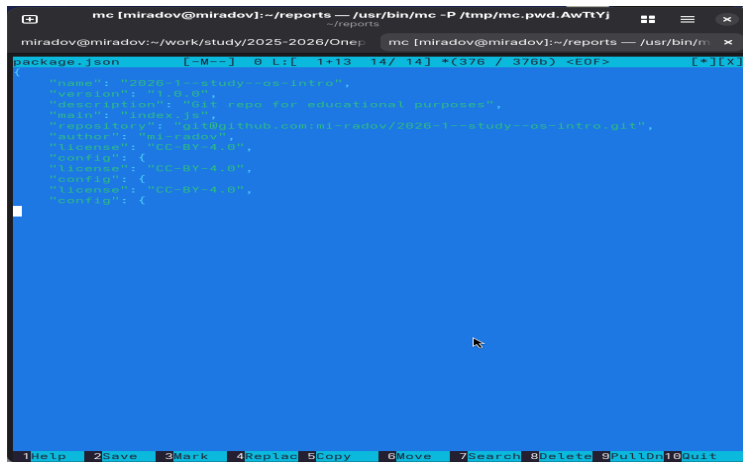


Рисунок 31: Переход в конец файла

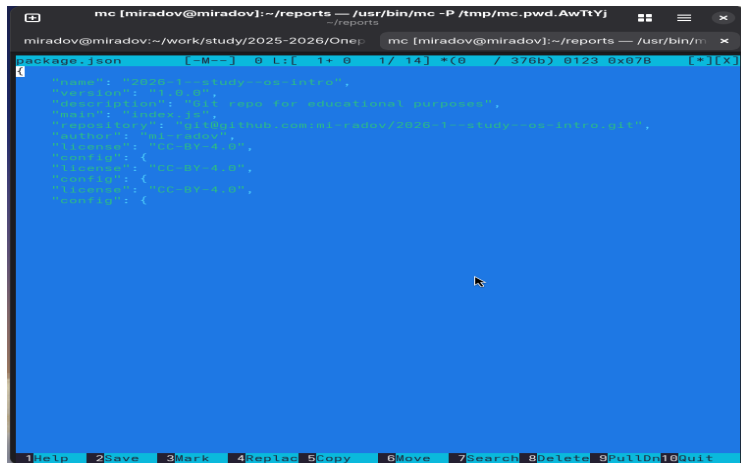
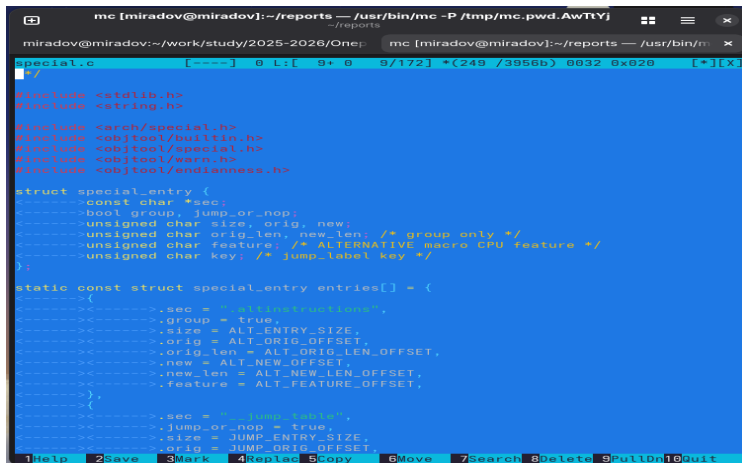


Рисунок 32: Переход в начало файла

Работа с редактором Midnight Commander



The screenshot shows the Midnight Commander (mc) file editor interface. The title bar indicates the current directory is `~/reports` and the file being edited is `special.c`. The editor displays the following C code:

```
special.c [-----] 0 L: [ 9+ 0 9/172] +(249 /3956b) 0032 0x020 [*][X]
#include <stdlib.h>
#include <string.h>

#include <arch/special.h>
#include <objtool/builtin.h>
#include <objtool/special.h>
#include <objtool/warn.h>
#include <objtool/endianness.h>

struct special_entry {
<----->const char *sec;
<----->bool group, jump_or_nop;
<----->unsigned char size, orig, new;
<----->unsigned char orig_len, new_len; /* group only */
<----->unsigned char feature; /* ALTERNATIVE macro CPU feature */
<----->unsigned char key; /* jump_label key */
};

static const struct special_entry entries[] = {
<----->{
<-----><----->.sec = "altinstructions",
<-----><----->.group = true,
<-----><----->.size = ALT_ENTRY_SIZE,
<-----><----->.orig = ALT_ORIG_OFFSET,
<-----><----->.orig_len = ALT_ORIG_LEN_OFFSET,
<-----><----->.new = ALT_NEW_OFFSET,
<-----><----->.new_len = ALT_NEW_LEN_OFFSET,
<-----><----->.feature = ALT_FEATURE_OFFSET,
<-----><----->},
<-----><----->.sec = "jump_label",
<-----><----->.jump_or_nop = true,
<-----><----->.size = JUMP_ENTRY_SIZE,
<-----><----->.orig = JUMP_ORIG_OFFSET,
<-----><----->},
<-----><----->};
```

The bottom status bar shows keyboard shortcuts: 1 Help, 2 Save, 3 Mark, 4 Replac, 5 Copy, 6 Move, 7 Search, 8 Delete, 9 PullDn, 10 Quit.

Рисунок 33: Файл с программой

Работа с редактором Midnight Commander

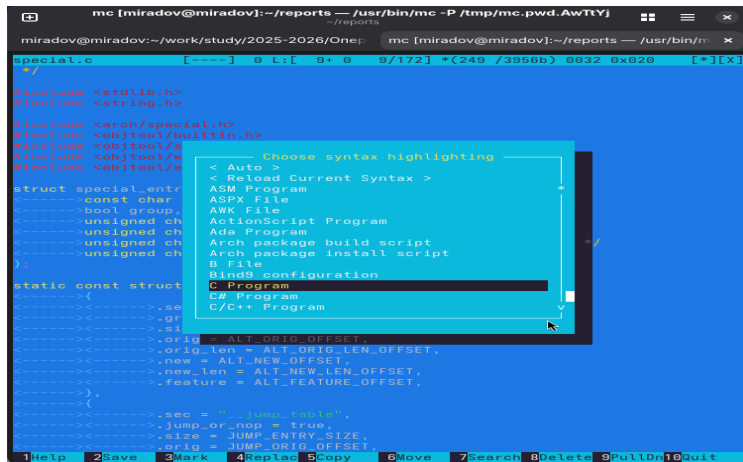


Рисунок 34: Цветовыделение синтаксиса

3. Выводы по проделанной работе

В данной работе мы ознакомились с инструментами поиска файлов и фильтрации текстовых данных. А также приобрели практические навыки по управлению процессами.